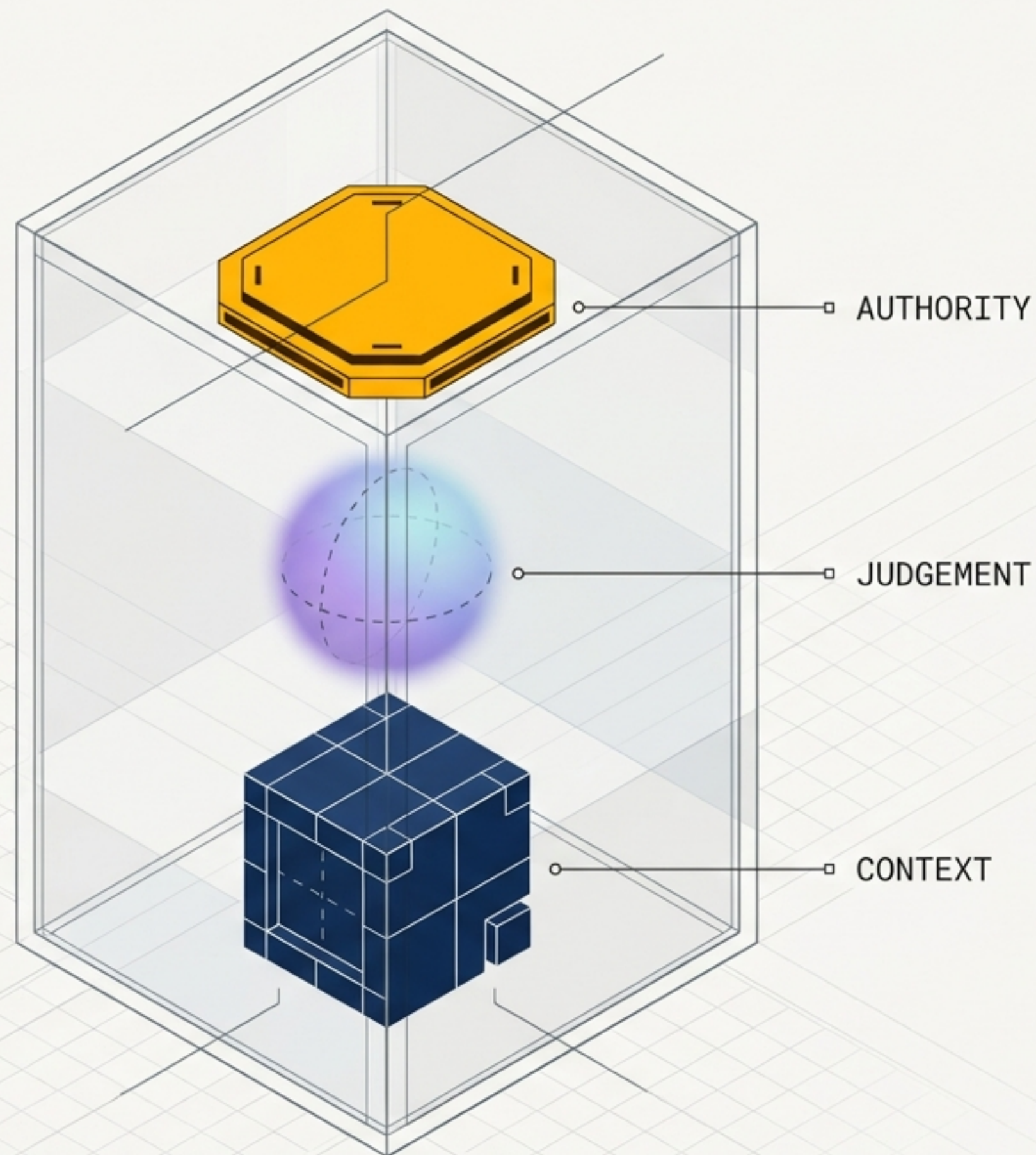


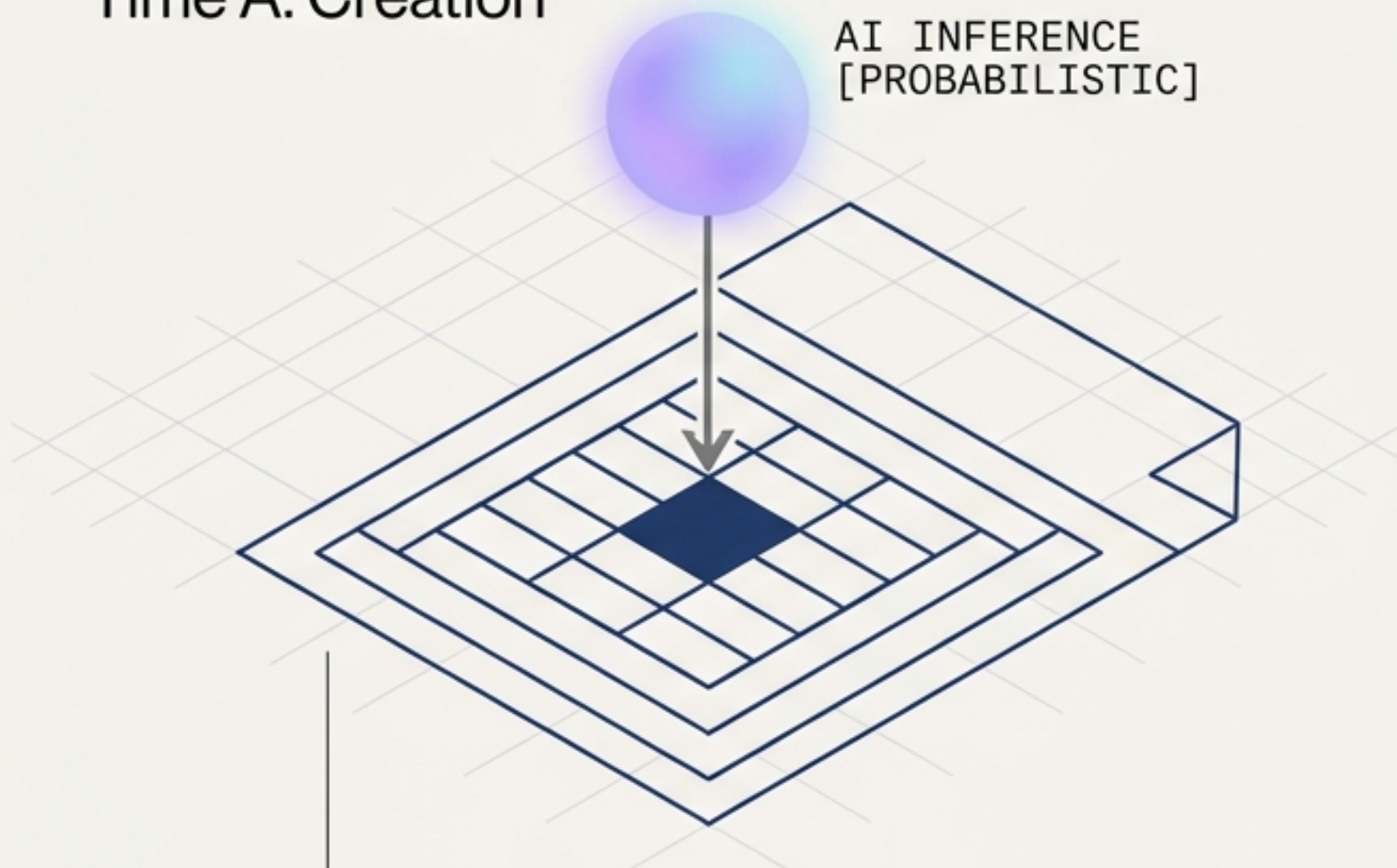
Accountable Intelligence

Bounding Probabilistic Judgement with
Persistent Context and Governed Action

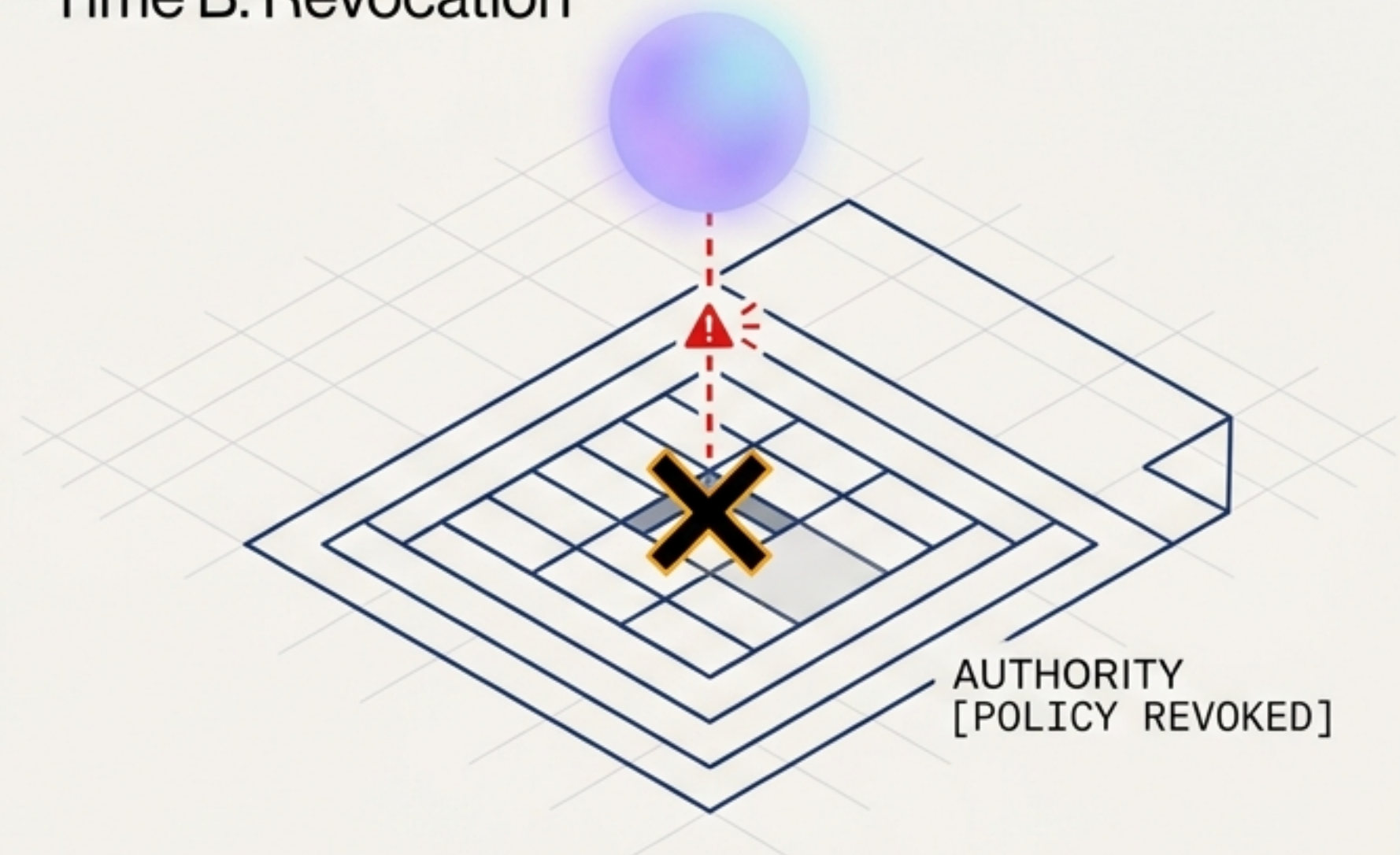


The Decision Needs a Context That Survives It

Time A: Creation



Time B: Revocation



A model can make a useful judgement and still leave the organization with an incomplete account.

- WHAT EVIDENCE WAS AVAILABLE AND WAS IT CURRENT?
- WHICH POLICY APPLIED AND WHO HAD AUTHORITY?
- WHAT HAPPENS IF A SUPPORTING ASSERTION IS LATER WITHDRAWN?

These questions persist after the inference finishes. They concern the continuity of organizational responsibility.

The Principle of Anti-Collapse

No single score should collapse these four distinct properties.

Structural Validity

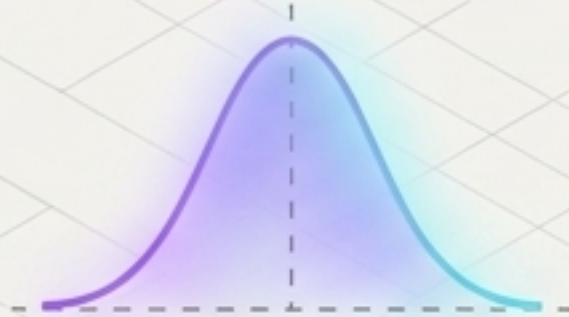


Does the result conform to the required schema?

A permitted category and a numeric score.

Crucially Does Not Establish:
That the selected category is correct.

Statistical Calibration



Do estimates match observed correctness?

Predictions near 0.9 succeed ~90% of the time.

Crucially Does Not Establish:
Certainty about one specific decision.

Context Verification



Which checks support this assertion?

Issuer, scope, and validity checks recorded.

Crucially Does Not Establish:
That every statement from the issuer is true.

Action Authority



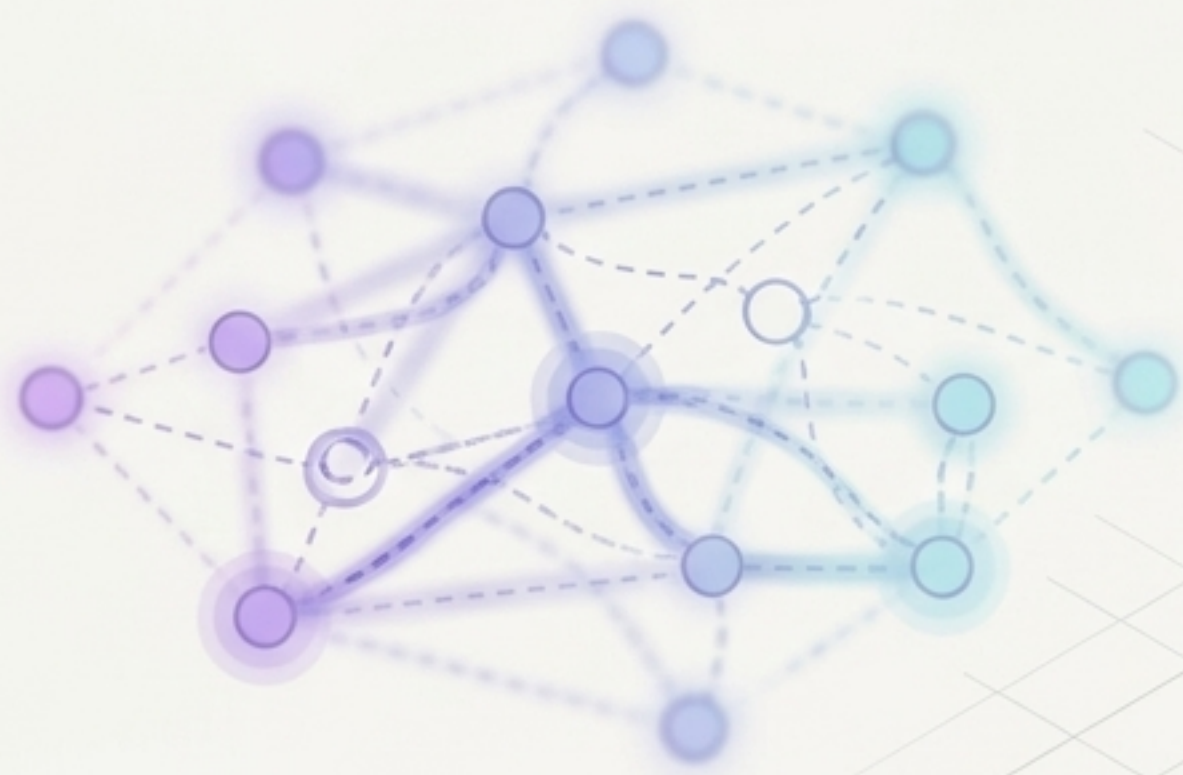
May this actor take this action?

A valid delegation permits operation.

Crucially Does Not Establish:
That the operation is wise or premise correct.

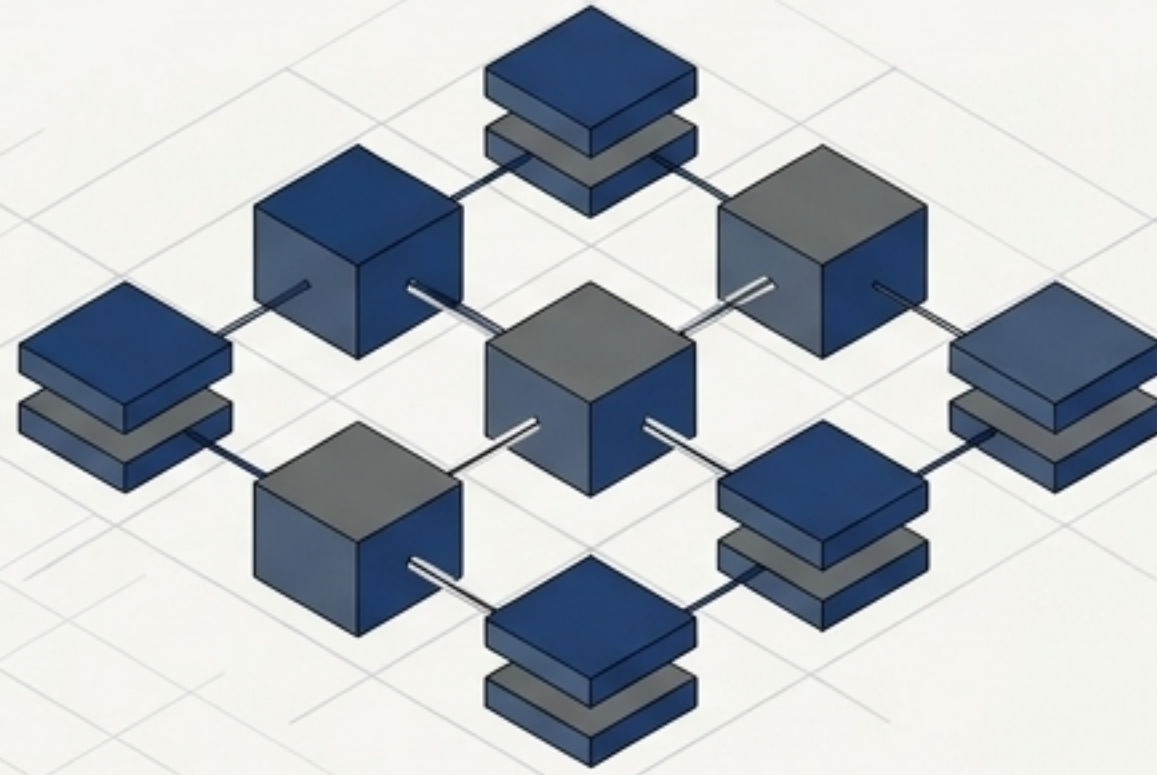
Separation of Concerns: Judgement vs. Context

TypeSafe AI (Jev Model)



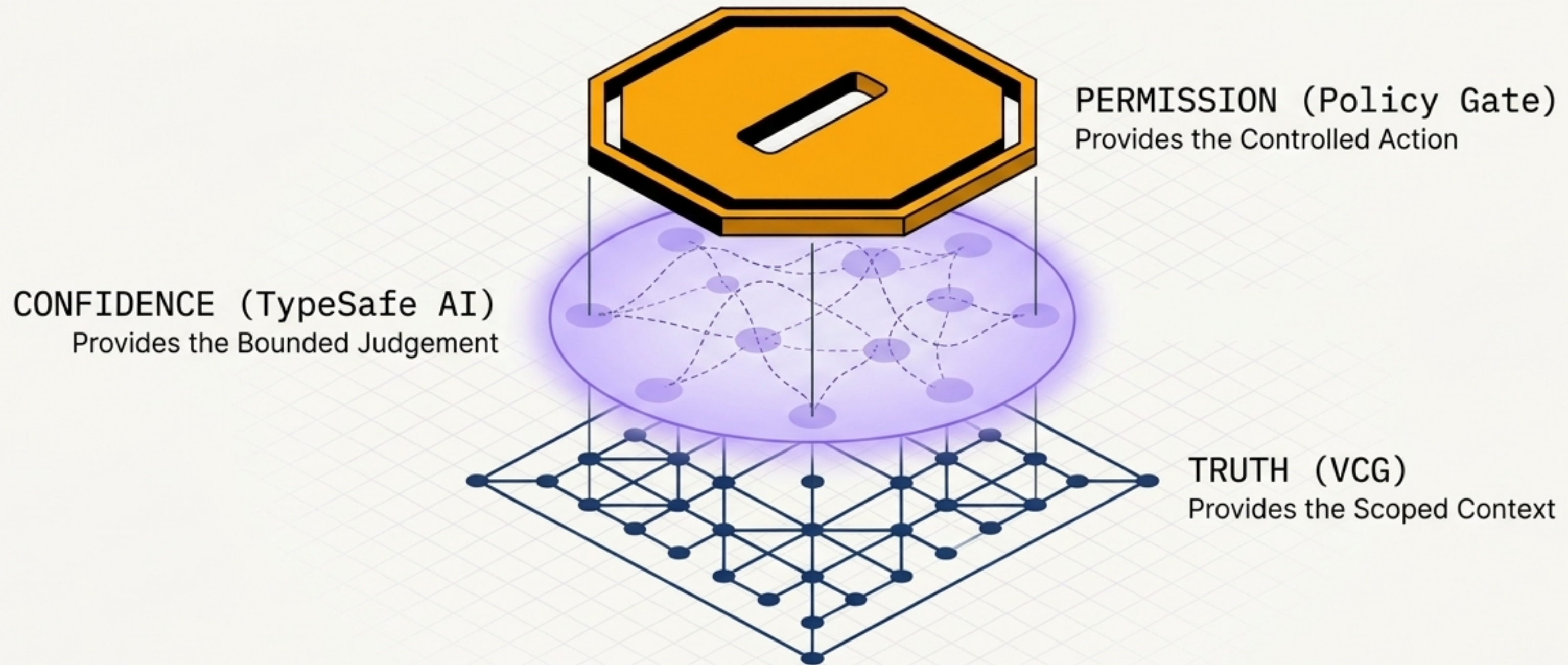
- Role: Supplies judgement where interpretation is required.
- Mechanism: System One Models using Reinforcement Learning for Calibrated Decisions.
- Output: Structured decisions, probabilities, and confidence estimates.

Verified Context Graphs (VCG)



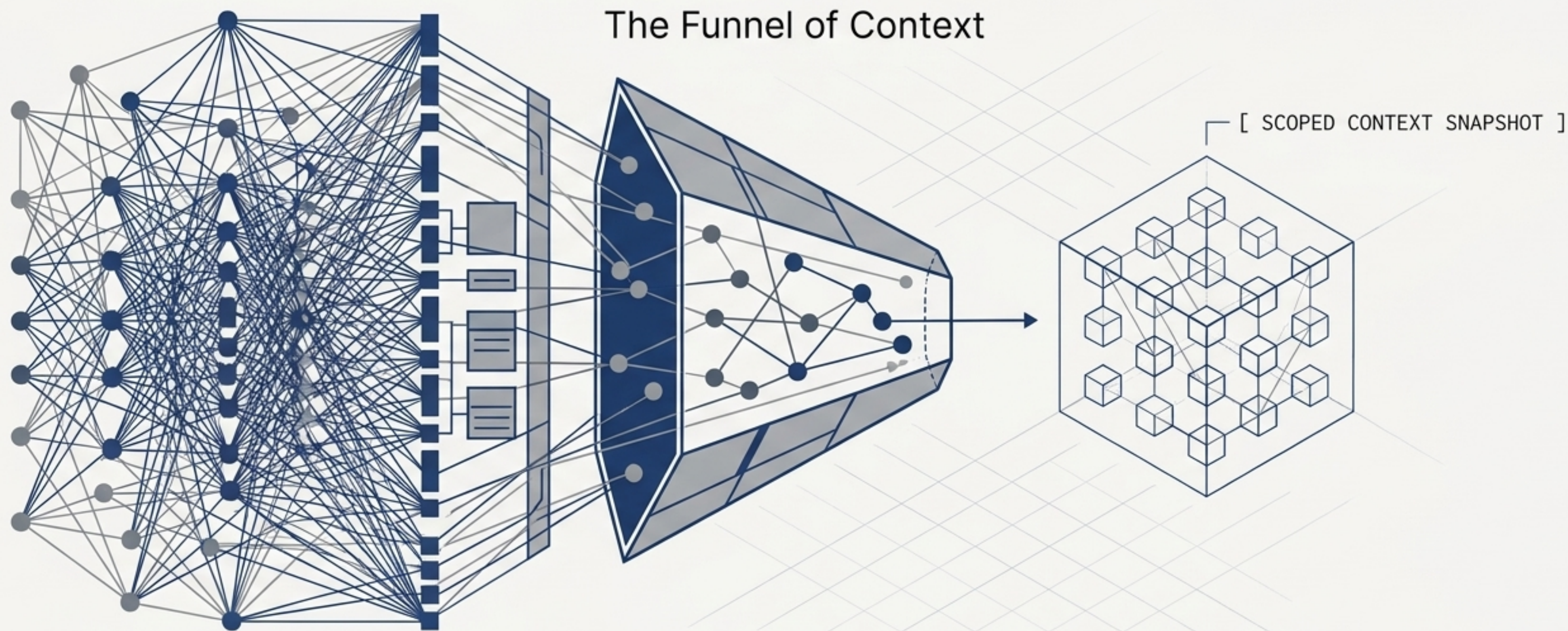
- Role: Preserves context, evidence relationships, and revision history.
- Mechanism: Checkable graph of statements, provenance, and authority.
- Output: Retains accepted information, missing evidence, and unresolved disagreements.

Confidence is Not Truth. Truth is Not Permission.



True accountable intelligence is achieved not by making AI universally perfect,
but by restricting its operational boundaries.

Step 1: Prepare a Bounded Context

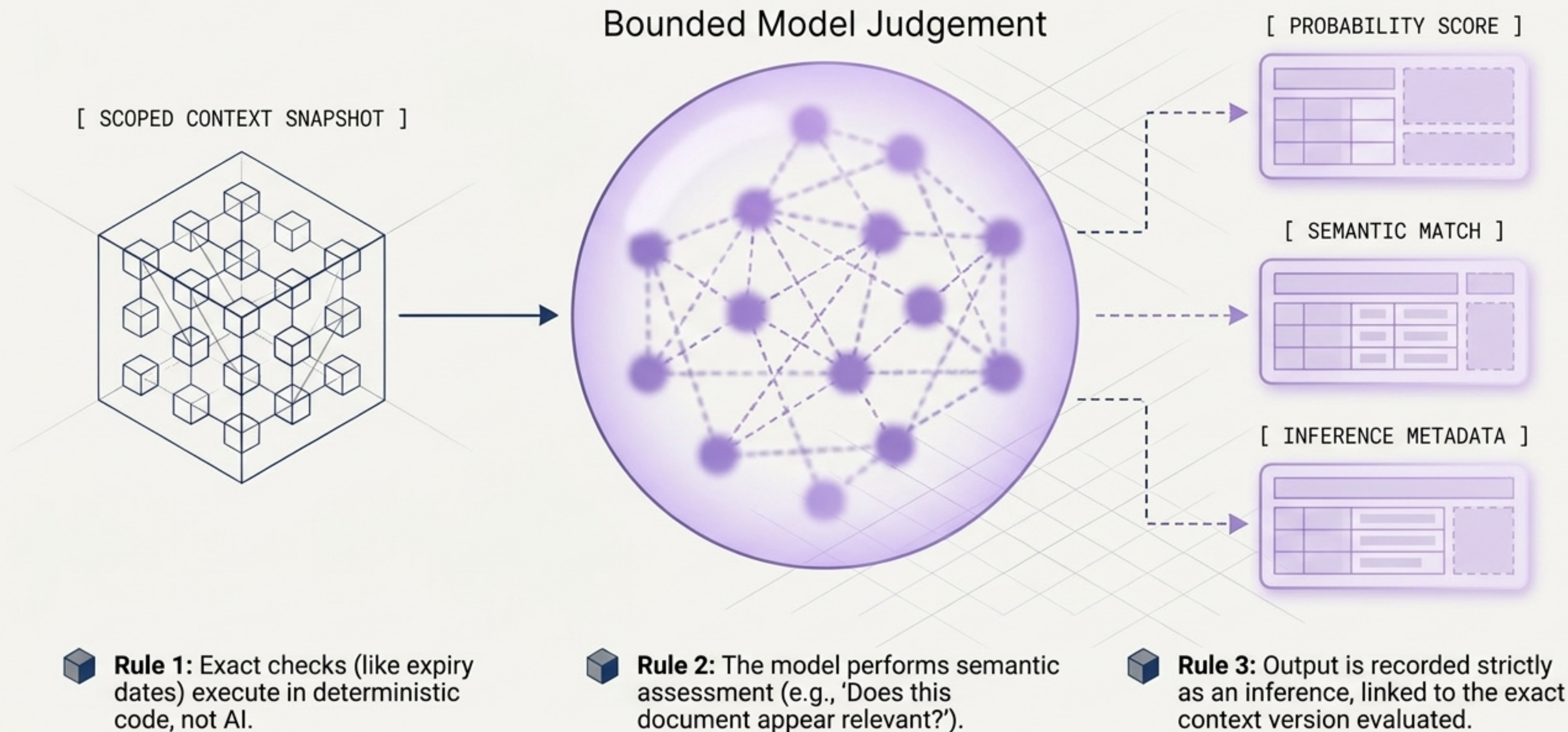


- **Deterministic query** selects relevant state (evidence references, validity intervals, unresolved conflicts).

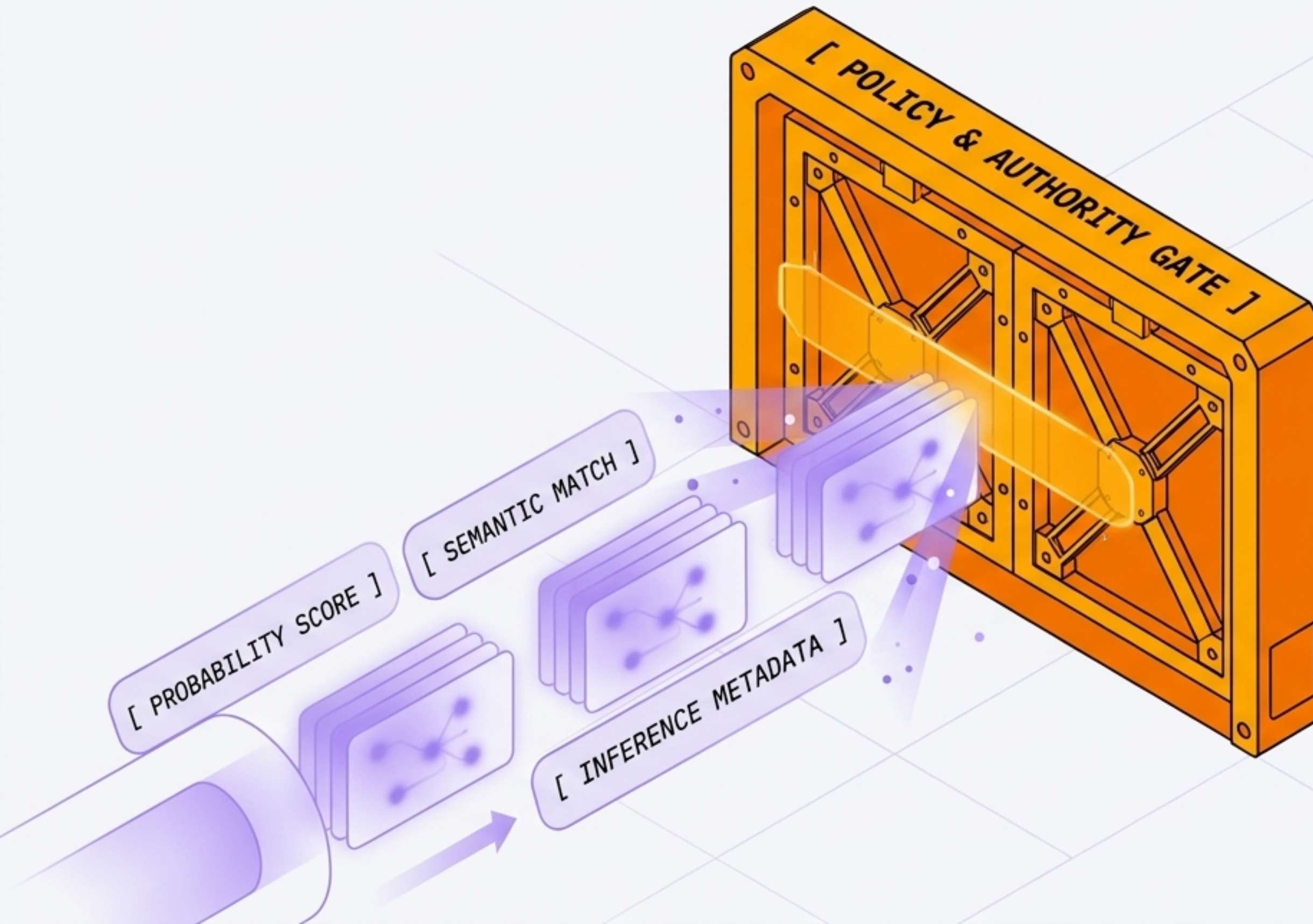
- Missing relationships appear explicitly as **unknowns**.

- Omission must never silently become a **negative finding**.

Step 2: Ask for the Judgement Actually Needed



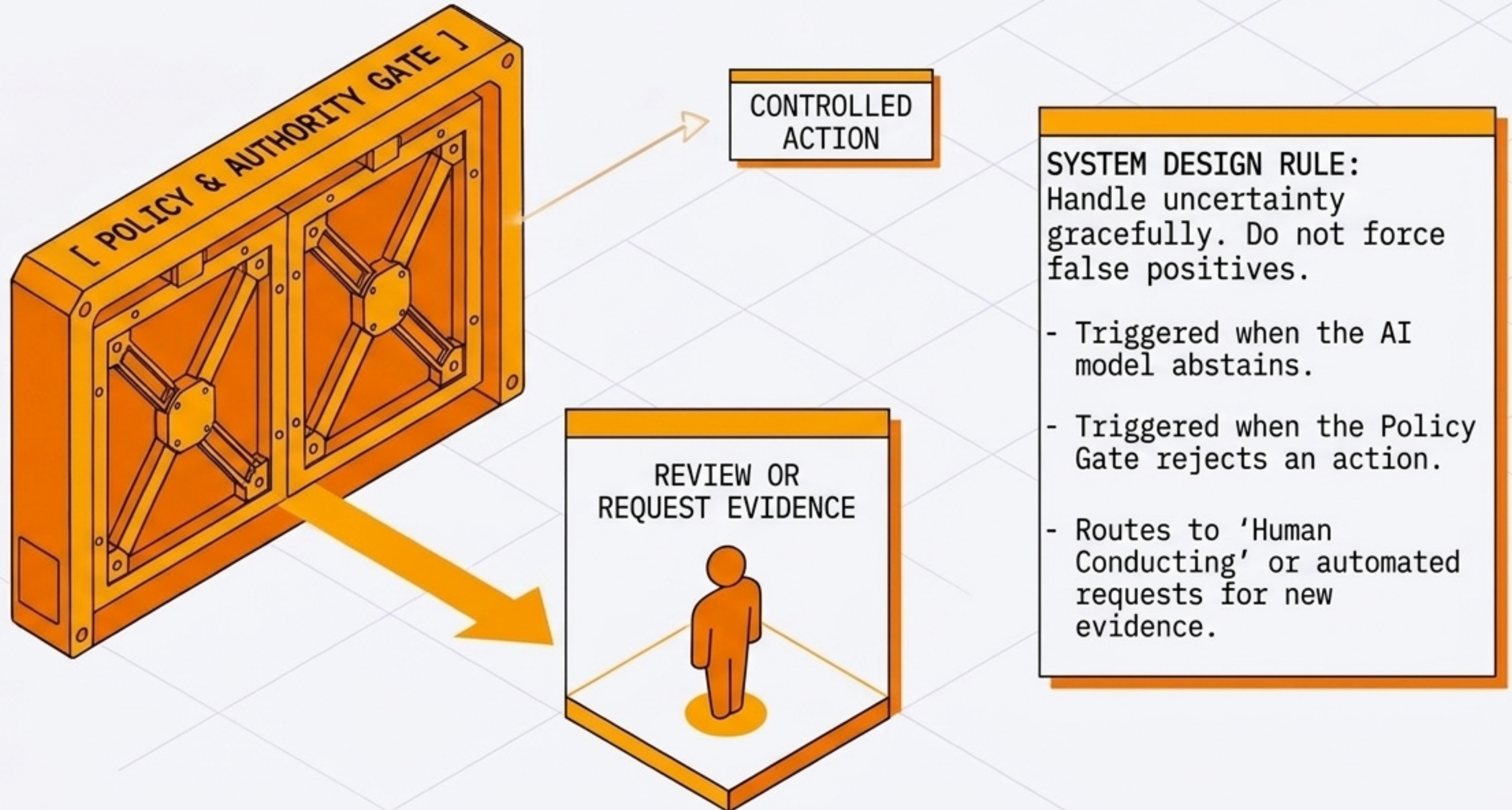
Step 3: Enforce the Action Boundary



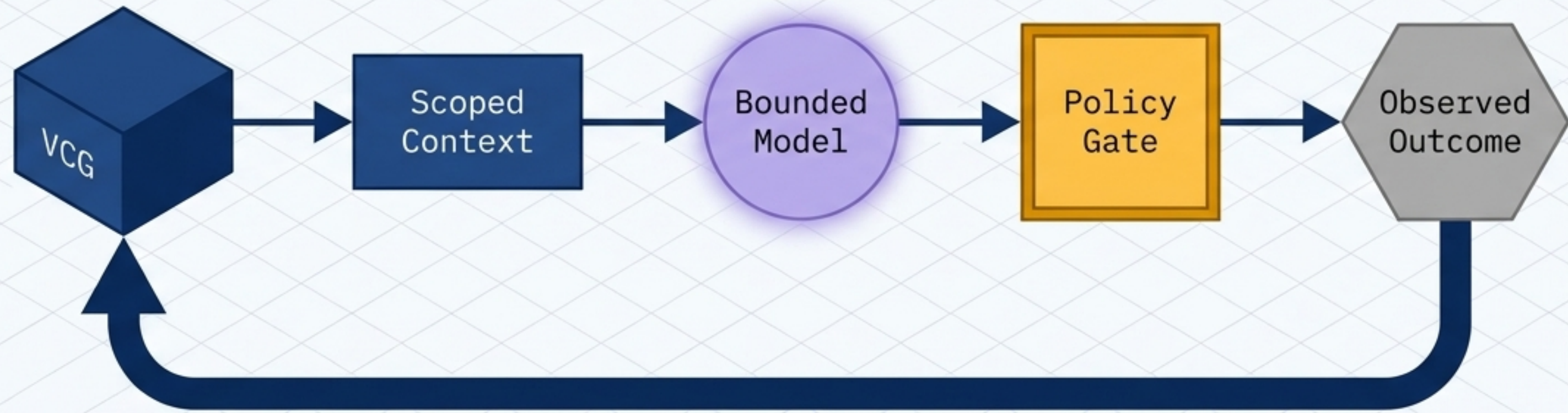
Data accessing flowponents from previous stipeline:

- Gate checks evidence requirements, delegations, and scanners, and policy constraints.
- Model confidence is merely an input; it cannot override failed authority checks.
- Gate detects material changes since the initial context cntext snapshot before allowing execution.

Step 4: The Unresolved Safety Valve



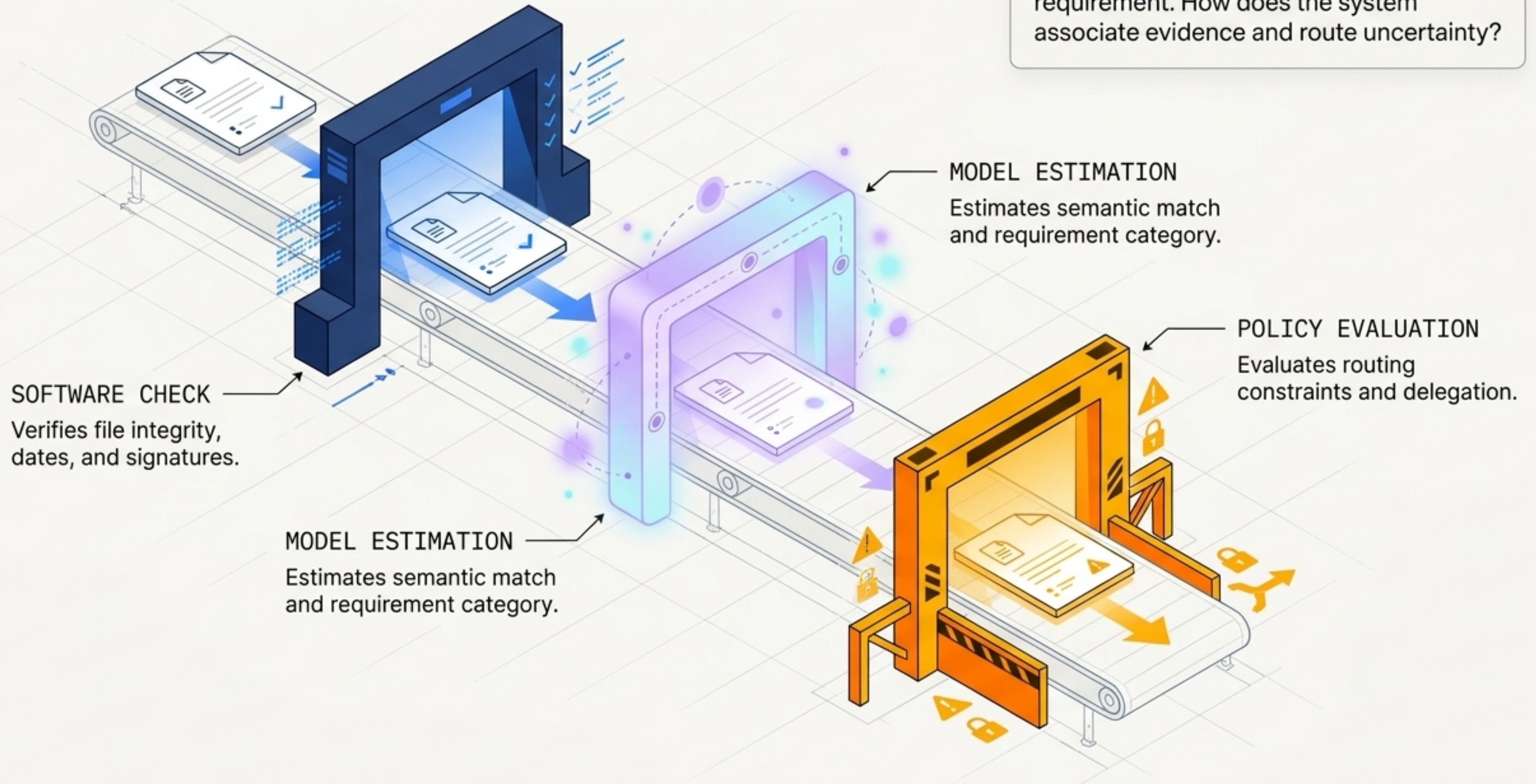
Step 5: Record the Consequence



- Proposed decision, permission result, attempted action, and observed outcome are recorded separately.
- Preserves the historical record while marking dependencies for future reassessment.
- Allows organizations to explain why a decision was reasonable at the time.

Application: Organisational Evidence Review

Scenario: An organization receives a document to satisfy an operational requirement. How does the system associate evidence and route uncertainty?



The Routing Decision Matrix

SITUATION	PROPOSED RESPONSE	ENFORCED BY
Relevant document + checks passed	Attach as supporting evidence	Policy Gate (Automated)
Relevant document + issuer authority unresolved	Request issuer verification	VCG (Context Gap)
Strong semantic match + expired validity	Record as historical, request new document	Deterministic Code
Conflicting scope / uncertain identity	Route to a qualified reviewer	Model (Uncertainty)
Evidence sufficient + actor lacks approval authority	Preserve assessment, refer approval	Policy Gate (Delegation)

Anatomy of an Accountable Record

Operational uncertainty requires more than a label and a probability. It requires a permanent, challengeable receipt.

QUESTION_AND_SCHEMA_VERSION

← What was judged.

MODEL_ID : JEV_v2
PROBABILITY : 0.89

← The AI's assessment.

CONTEXT_SNAPSHOT_REF
VCG_VERIFICATION_STATUS : PASSED

← Condition of the evidence.

POLICY_VERSION_APPLIED
AUTHORITY_REFERENCE_ID

← The permission.

EXPIRY_DATA
ACTION_RECEIPT

← Intention vs. consequence.

System Anti-Patterns: The Failure Modes

PROMOTION OF INFERENCE

A fuzzy inference is stored without a tag, eventually treated as a verified hard fact.



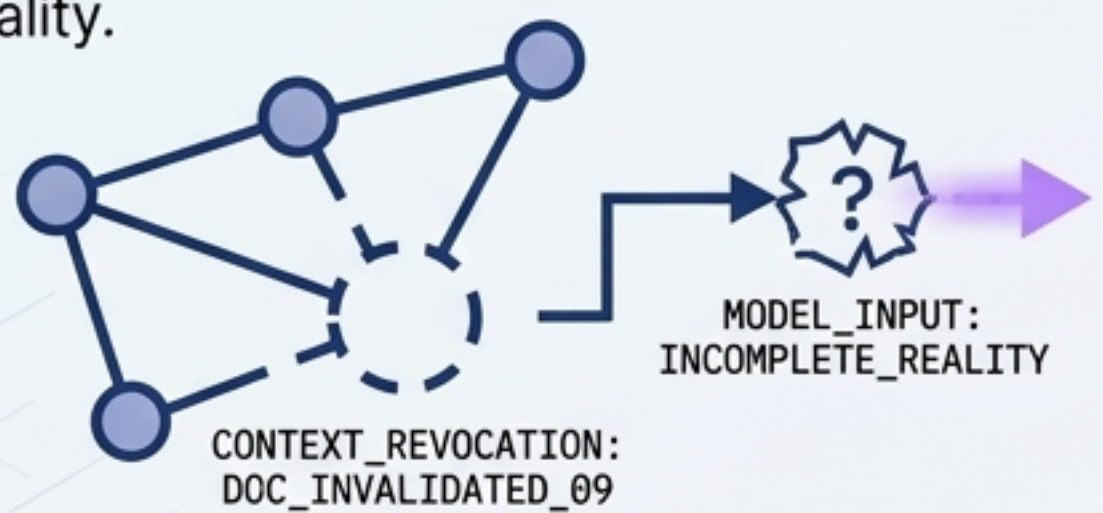
AUTHORITY CONFUSION

Software misinterprets a model finding a document 'plausible' as explicit permission to act.



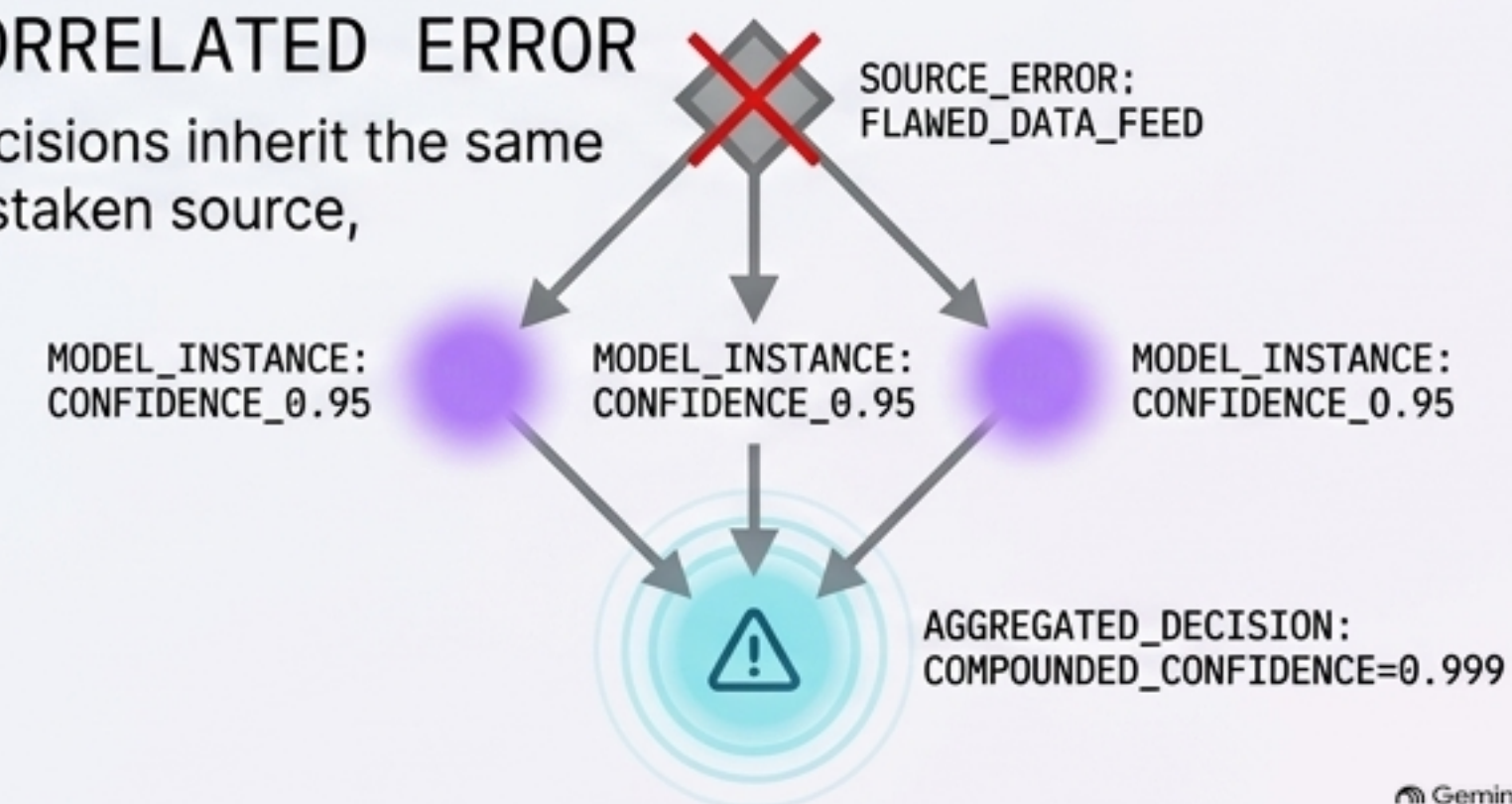
SELECTIVE CONTEXT

Retrieval omits a critical revocation, giving the model an incomplete reality.

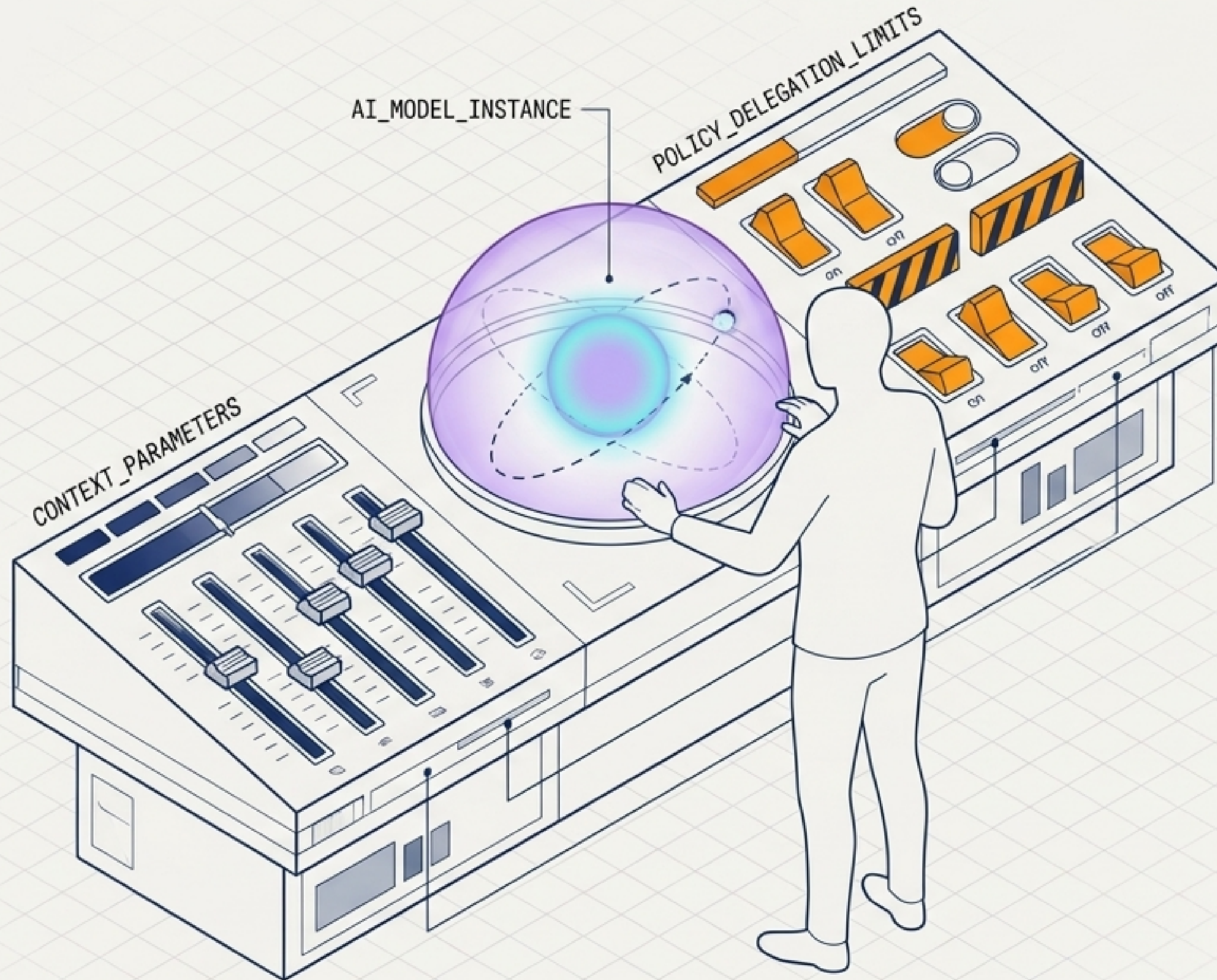


CORRELATED ERROR

Decisions inherit the same mistaken source,



The Human Element: Conducting



- This architecture changes where human attention is most valuable.
- **CONDUCTING:** Holding direction and coordinating differentiated capabilities.
- Humans establish objectives, define acceptable evidence, and dictate delegated consequences.
- Maintains robust routes for challenge and correction.
- **Benefit:** Reduces repetitive interpretation while preserving total accountability.

The Evaluation Framework

Testing through narrow, reversible workflows before extending autonomy.

Testing Scorecard	
Decision Quality	Task accuracy & consequence-weighted errors.
Uncertainty Quality	Reliability diagrams & errors at automation thresholds.
Context Quality	Missing prerequisites & incorrect entity associations.
Authority Enforcement	Do expired delegations block prohibited transitions?
Change Handling	Do revocations trigger automatic reassessment?
Human Accountability	The ability to explain, challenge, and correct a consequential decision.

The Mandate for Accountable Intelligence

